

Experiment: Data Preprocessing Using Pandas in Google Colab

Experiment Title

Data Preprocessing and Cleaning of Student Performance Dataset Using Pandas in Google Colab

Aim

To perform data preprocessing and cleaning operations on a real-world dataset using Pandas in Google Colab.

Learning Outcomes

1. Upload and import datasets in Google Colab.
2. Explore and understand dataset structure using Pandas.
3. Handle missing values and duplicate records.
4. Clean and transform categorical and numerical data.
5. Create new features from existing data.
6. Export cleaned datasets for further analysis.

Software and Platform Requirements

- Google Account
- Internet Connection
- Google Colab
- Pandas

Dataset Information

Dataset Name: Student Performance Dataset

Dataset Source:

<https://www.kaggle.com/datasets/spscientist/students-performance-in-exams>

Dataset File Required:

StudentsPerformance.csv

Steps to Download the Dataset

1. Open the dataset link.
2. Sign in to Kaggle.
3. Click the Download button.
4. Extract the downloaded ZIP file.
5. Locate the file named StudentsPerformance.csv.
6. Save this CSV file on your computer.

Steps to Open Google Colab

1. Open <https://colab.research.google.com>
2. Sign in using your Google account.
3. Click New Notebook.
4. Rename the notebook appropriately.

Experiment Structure

Experiment Folder

```
|  
├── StudentsPerformance.csv  
├── preprocessing_experiment.ipynb  
└── cleaned_student_dataset.csv
```

Theory

Data preprocessing is the process of cleaning and transforming raw data into a structured and usable format. Real-world datasets may contain duplicate entries, inconsistent formatting, categorical data, or missing values. Pandas provides efficient tools to preprocess such datasets before analysis or machine learning.

Task 1: Upload Dataset in Google Colab

Objective: Upload the dataset file into the Colab environment.

Instructions:

1. Open the Google Colab notebook.
2. Locate the Files section on the left sidebar.
3. Click the Upload button.
4. Select the file StudentsPerformance.csv.
5. Verify that the file appears in the Colab file list.

Task 2: Import Required Library

Objective: Prepare the environment for preprocessing.

Instructions:

1. Import the Pandas library.
2. Verify successful import without errors.

Task 3: Load the Dataset

Objective: Read the CSV dataset into a Pandas DataFrame.

Instructions:

1. Load the CSV file from the Colab environment.
2. Store the dataset in a DataFrame.
3. Display the first few records.
4. Observe rows and columns carefully.

Task 4: Explore Dataset Information

Objective: Understand the structure of the dataset.

Instructions:

1. Display column names.
2. Check the number of rows and columns.
3. Observe data types of each column.
4. Check non-null value counts.
5. Generate summary statistics for numerical columns.

Task 5: Identify Missing Values

Objective: Detect incomplete or unavailable data.

Instructions:

1. Check whether missing values exist.
2. Count missing values column-wise.
3. Identify columns containing null values.
4. Analyze the impact of missing data.

Task 6: Handle Missing Values

Objective: Clean the dataset by treating missing entries.

Instructions:

1. Replace missing categorical values with suitable labels.
2. Handle missing numerical values appropriately.
3. Verify that missing values are removed or replaced.
4. Recheck null counts after preprocessing.

Task 7: Remove Duplicate Records

Objective: Ensure consistency in the dataset.

Instructions:

1. Identify duplicate rows.
2. Count duplicate entries.
3. Remove duplicate records.
4. Verify successful removal.

Task 8: Rename and Clean Columns

Objective: Improve readability and consistency.

Instructions:

1. Observe existing column names.
2. Rename columns into simpler formats if required.

3. Remove unnecessary spaces from column names.
4. Verify updated column names.

Task 9: Encode Categorical Data

Objective: Convert categorical information into usable numerical form.

Instructions:

1. Identify categorical columns such as Gender, Lunch, and Test Preparation Course.
2. Convert categorical labels into numerical representation.
3. Verify successful conversion.

Task 10: Create New Features

Objective: Generate useful information from existing data.

Instructions:

1. Create a column for total marks.
2. Create a column for average marks.
3. Categorize students into performance groups such as Excellent, Good, Average, and Poor.
4. Verify newly created columns.

Task 11: Filter and Analyze Data

Objective: Perform basic analysis using preprocessing techniques.

Instructions:

1. Display students scoring above a chosen threshold.
2. Filter records based on gender.
3. Find average marks for each category.
4. Compare student performance across groups.
5. Identify highest and lowest scoring students.

Task 12: Save the Cleaned Dataset

Objective: Export the processed dataset.

Instructions:

1. Save the cleaned DataFrame as a new CSV file.
2. Name the file appropriately.
3. Verify that the file appears in the Colab file section.
4. Download the cleaned CSV file to the local system.

Expected Output

- A cleaned and processed dataset
- Removed duplicate records
- Handled missing values
- Encoded categorical data

- Newly generated feature columns
- A downloadable cleaned CSV dataset

Result

The Student Performance dataset was successfully preprocessed using Pandas in Google Colab. Missing values and duplicate records were handled, categorical data was encoded, and new analytical features were created successfully.

Viva Questions

1. What is data preprocessing?
2. Why is preprocessing important in data analysis?
3. What is a DataFrame?
4. What are missing values?
5. Why are duplicate records removed?
6. What is categorical encoding?
7. What is feature engineering?
8. Why is Google Colab useful for data analysis?
9. What is the purpose of exporting cleaned data?
10. What are the advantages of Pandas?